```
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

====================== RESTART: D:/machine learnings.py ======================
Warning: TA library not available. Technical indicators will be limited.
🚀 NIFTY 50 Advanced Trading System
==================================================
📥 Loading NIFTY data...
📄 Processing: NIFTY 50-01-01-2020-to-01-01-2021.csv
📋 Detected columns:
  date: 'Date '
  open: 'Open '
  high: 'High '
  low: 'Low '
  close: 'Close '
  volume: 'Shares Traded '
  turnover: 'Turnover (₹ Cr)'
✅ Successfully processed NIFTY 50-01-01-2020-to-01-01-2021.csv
📄 Processing: NIFTY 50-01-01-2021-to-01-01-2022.csv
📋 Detected columns:
  date: 'Date '
  open: 'Open '
  high: 'High '
  low: 'Low '
  close: 'Close '
  volume: 'Shares Traded '
  turnover: 'Turnover (₹ Cr)'
✅ Successfully processed NIFTY 50-01-01-2021-to-01-01-2022.csv
📄 Processing: NIFTY 50-01-01-2022-to-01-01-2023.csv
📋 Detected columns:
  date: 'Date '
  open: 'Open '
  high: 'High '
  low: 'Low '
  close: 'Close '
  volume: 'Shares Traded '
  turnover: 'Turnover (₹ Cr)'
✅ Successfully processed NIFTY 50-01-01-2022-to-01-01-2023.csv
📄 Processing: NIFTY 50-01-01-2023-to-01-01-2024.csv
📋 Detected columns:
  date: 'Date '
  open: 'Open '
  high: 'High '
  low: 'Low '
  close: 'Close '
  volume: 'Shares Traded '
  turnover: 'Turnover (₹ Cr)'
✅ Successfully processed NIFTY 50-01-01-2023-to-01-01-2024.csv
📄 Processing: NIFTY 50-01-01-2024-to-01-01-2025.csv
```

```
📋 Detected columns:
  date: 'Date '
  open: 'Open '
  high: 'High '
  low: 'Low '
  close: 'Close '
  volume: 'Shares Traded '
  turnover: 'Turnover (₹ Cr)'
✅ Successfully processed NIFTY 50-01-01-2024-to-01-01-2025.csv

🎯 Data Summary:
  📅 Date Range: 2020-01-01 00:00:00 to 2025-01-01 00:00:00
  📊 Total Records: 1244
  📈 Available Columns: ['Date', 'Open', 'High', 'Low', 'Close', 'Volume',
'Turnover', 'SourceFile']

🔧 Adding advanced technical features...
✅ Added 74 total features

🎯 Creating prediction targets...
✅ Created 18 different prediction targets

🧹 Preparing features and targets...
📊 Feature selection: 64 features out of 68 candidates
✅ Final feature matrix shape: (1244, 64)

============================================================
TRAINING MODELS FOR: Direction_1d
============================================================

🤖 Training models for target: Direction_1d
📊 Data split: Train=995, Test=249
📈 Target distribution - Train: {1: 552, 0: 443}
📈 Target distribution - Test: {1: 137, 0: 112}
🔄 Training Random Forest...
✅ Random Forest: Accuracy=0.462, F1=0.562
🔄 Training Gradient Boosting...
✅ Gradient Boosting: Accuracy=0.478, F1=0.500
🔄 Training Logistic Regression...
✅ Logistic Regression: Accuracy=0.446, F1=0.138
✅ Ensemble: Accuracy=0.478, F1=0.435

📊 Creating visualizations for Direction_1d...
✅ Visualizations saved to: D:\Users\TANISH\Downloads\New folder\advanced_plots

📈 Backtesting strategy for Direction_1d...
🏆 Using best model: Random Forest

📊 Backtest Results Summary:
  💰 Initial Capital: ₹100,000
```

```
  💰 Final Capital: ₹108,491.85
  📈 Total Return: 8.49%
  📊 Benchmark Return: 9.59%
  🎯 Excess Return: -1.10%
  📉 Sharpe Ratio: 0.89
  📉 Max Drawdown: 10.93%
  🎯 Win Rate: 0.00%
  📊 Total Trades: 1
💾 Model saved to: D:\Users\TANISH\Downloads\New
folder\advanced_plots\nifty_model_Direction_1d_20250902_111323.pkl

============================================================
TRAINING MODELS FOR: Direction_3d
============================================================

🤖 Training models for target: Direction_3d
📊 Data split: Train=995, Test=249
📈 Target distribution - Train: {1: 570, 0: 425}
📈 Target distribution - Test: {1: 145, 0: 104}
🔄 Training Random Forest...
✅ Random Forest: Accuracy=0.466, F1=0.543
🔄 Training Gradient Boosting...
✅ Gradient Boosting: Accuracy=0.402, F1=0.361
🔄 Training Logistic Regression...
✅ Logistic Regression: Accuracy=0.442, F1=0.388
✅ Ensemble: Accuracy=0.418, F1=0.388

📊 Creating visualizations for Direction_3d...
✅ Visualizations saved to: D:\Users\TANISH\Downloads\New folder\advanced_plots

📈 Backtesting strategy for Direction_3d...
🏆 Using best model: Random Forest

📊 Backtest Results Summary:
  💰 Initial Capital: ₹100,000
  💰 Final Capital: ₹108,825.28
  📈 Total Return: 8.83%
  📊 Benchmark Return: 9.59%
  🎯 Excess Return: -0.76%
  📉 Sharpe Ratio: 0.74
  📉 Max Drawdown: 10.93%
  🎯 Win Rate: 0.00%
  📊 Total Trades: 1
💾 Model saved to: D:\Users\TANISH\Downloads\New
folder\advanced_plots\nifty_model_Direction_3d_20250902_111348.pkl

============================================================
TRAINING MODELS FOR: Significant_Up_1d
============================================================
```

```
🤖 Training models for target: Significant_Up_1d
📊 Data split: Train=995, Test=249
📈 Target distribution - Train: {0: 850, 1: 145}
📈 Target distribution - Test: {0: 232, 1: 17}
🔄 Training Random Forest...
✅ Random Forest: Accuracy=0.932, F1=0.000
🔄 Training Gradient Boosting...
✅ Gradient Boosting: Accuracy=0.932, F1=0.000
🔄 Training Logistic Regression...
✅ Logistic Regression: Accuracy=0.932, F1=0.190
✅ Ensemble: Accuracy=0.932, F1=0.000

📊 Creating visualizations for Significant_Up_1d...
✅ Visualizations saved to: D:\Users\TANISH\Downloads\New folder\advanced_plots

📈 Backtesting strategy for Significant_Up_1d...
🏆 Using best model: Logistic Regression

📊 Backtest Results Summary:
  💰 Initial Capital: ₹100,000
  💰 Final Capital: ₹108,491.85
  📈 Total Return: 8.49%
  📊 Benchmark Return: 9.59%
  🎯 Excess Return: -1.10%
  📉 Sharpe Ratio: 0.89
  📉 Max Drawdown: 10.93%
  🎯 Win Rate: 0.00%
  📊 Total Trades: 1
💾 Model saved to: D:\Users\TANISH\Downloads\New
folder\advanced_plots\nifty_model_Significant_Up_1d_20250902_111415.pkl

📝 Generating comprehensive report...
📄 Report saved to: D:\Users\TANISH\Downloads\New
folder\advanced_plots\NIFTY_Analysis_Report.txt

🎉 Analysis complete!
📁 All results saved to: D:\Users\TANISH\Downloads\New folder\advanced_plots
📊 Check the plots and report for detailed insights.
```